

Introduction to the Evaluation Framework

Human computer interaction (HCI) is the design, evaluation and implementation of interactive systems for human use. Websites are a primary way people access news, entertainment and sport, making evaluation of their design quality essential to ensure a good user experience. Poor design can frustrate users and damage trust. This framework provides a structured approach to evaluating websites across three sectors, news, entertainment and sport by translating established HCI and design principles into measurable criteria.

Usability Principles

Usability is defined as how a product is used by specified users to achieve goals with effectiveness, efficiency and satisfaction [1]. In website design, this refers to whether users can complete tasks without unnecessary difficulty or errors. A widely used evaluation method is Jakob Nielsen's ten usability heuristics, covering principles such as system visibility, error prevention, consistency and recognition rather than recall [2]. Don Norman's *The Design of Everyday Things* introduces concepts such as affordances, signifiers and feedback, assessing whether elements clearly communicate their function [3]. Together, these provide the theoretical basis for the evaluations in this report.

Accessibility

Accessibility ensures websites can be used by as many people as possible, including those with visual, motor, cognitive or hearing impairments. The Web Content Accessibility Guidelines (WCAG), published by the W3C, are the international standard, structured around four principles: perceivable, operable, understandable and robust [4]. Key considerations include contrast ratios, readable text, alternative text for images, full keyboard navigation and screen reader compatibility through semantic HTML and ARIA labels. Accessibility reflects inclusive design, ensuring no users are excluded. Evaluation will involve keyboard navigation testing and contrast checks.

Visual Design and Hierarchy

Visual design refers to how information is presented, while visual hierarchy organises elements to guide user attention based on priority. Gestalt principles—particularly proximity, similarity and continuity—explain how users group visual information, meaning consistent design implies consistent function [5]. Typography, including font size and spacing, strongly affects readability. Images and infographics should support content rather than distract. Evaluation will assess clarity of hierarchy, consistency of typography and whether visuals enhance usability.

Colour Theory and Use of Colour

Colour strongly influences user perception, attention and readability. Research shows colours carry associations, such as blue for trust, red for urgency and green for success. These can reinforce communication but must be used carefully, as relying on colour alone creates accessibility barriers. WCAG specifies minimum contrast ratios and consistent use across a site [4]. Evaluation will check contrast, alignment with site purpose and whether meaning is communicated beyond colour alone.

Semiotics

Semiotics studies how signs convey meaning. In interface design, icons, indexes and symbols communicate functionality. Common icons, such as a magnifying glass for search or an envelope for email, are widely recognised, while unclear icons can confuse users. Colour also carries meaning, such as red for errors and green for success. Inconsistent use can reduce usability. Evaluation will assess clarity and consistency of icons, symbols and colour conventions.

Navigation and Information Architecture

Navigation and information architecture (IA) determine how users move through a website and locate content. Effective IA supports both browsing and goal-directed searching. Poor organisation or excessive choices can overwhelm users and slow task completion. Consistent navigation placement reduces learning effort. Evaluation will involve locating content using menus, assessing labelling and grouping, and testing navigation consistency and search functionality.

User-Centred Design (UCD)

User-centred design is an approach that focuses on the needs, goals and limitations of the websites intended users throughout the design process. Instead of designing purely around the business objective, UCD encourages designers to understand who their users are and what they are trying to achieve. A commonly used tool in this process is the persona, which is a fictional but research-based representation of a target user, outlining their goals, behaviours and any frustrations they may have as well [7]. UCD also involves a process where designs are prototyped by testing with real users and then improved based on feedback. User-centred design can be seen in a finished website by how well it suits user's needs, how easily it supports common tasks and whether it feels like its designed for users rather than just for advertising or brand promotion. Evaluation on will be based on this and how effectively each website serves its intended audience.

Cognitive Factors and Mental Workload

Cognitive factors influence how users process information. Cognitive load theory suggests working memory is limited, and exceeding this leads to errors and frustration [6]. Users can typically hold around seven items at once, influencing menu design and

content grouping [7]. Cluttered interfaces increase mental effort, while too many choices slow decision-making. Nielsen's recognition over recall principle supports presenting options clearly [2]. Evaluation will consider page complexity, number of choices and how well unnecessary effort is reduced.

Interaction and User Feedback

Effective interaction design ensures systems communicate clearly with users by confirming actions, indicating loading states and providing meaningful error messages. Norman identifies feedback as a key design principle, with its absence leading to frustration [3]. Examples include button states, loading indicators and clear error messages. Schneiderman's rules emphasise error recovery, allowing users to fix issues without losing progress [8]. Evaluation will test interactive elements and assess clarity of system feedback.

Ethical Design Considerations

Ethical design considers whether interfaces treat users honestly and support informed decision-making. Commercial pressures can prioritise organisational goals over user needs. Issues include misleading layouts and unclear consent mechanisms. The ACM Code of Ethics emphasises avoiding harm and ensuring transparency, while value-sensitive design integrates ethics into the design process [9]. Evaluation will assess transparency and whether design choices mislead users.

Dark Patterns and Persuasive Design

Dark patterns are design choices that manipulate users into unintended actions, often exploiting cognitive biases. Identified by Harry Brignull, examples include roach motels, confirm shaming and false urgency [10]. The key issue is when persuasion benefits organisations at the user's expense. Evaluation will identify hidden costs, misleading urgency and unclear opt-in or opt-out mechanisms.

Privacy and Security Considerations

Privacy concerns how websites collect, use and store data, and whether users are informed and in control. The UK GDPR requires transparency, explicit consent and data rights [11]. From a design perspective, clear communication through HTTPS, privacy policies and cookie controls builds trust. However, cookie interfaces often use dark patterns to encourage acceptance. Evaluation will check security indicators, privacy clarity and consent mechanisms.

User Support and Help Systems

User support is essential when users encounter issues. Nielsen's heuristics state that help should be easy to find, task-focused and clearly written [2]. Effective support includes contextual guidance, FAQs and chatbots where appropriate. Error messages

should explain what went wrong and how to fix it. Evaluation will involve locating help features and testing error recovery.

Framework design justification

The thirteen criteria cover key HCI and design principles while remaining practical to apply, forming a consistent checklist for comparing websites across three sectors. Usability and accessibility are fundamental, as without them users may struggle to complete tasks or access content. Visual design, colour theory and semiotics address how information is presented, while navigation and information architecture ensure it is clearly structured and accessible. User-centred design, cognitive factors, and interaction evaluate how well websites support users and minimise effort. Ethical design, dark patterns, privacy and user support emphasise trust and responsible design, particularly in commercial contexts, ensuring a balanced and comprehensive evaluation across all websites.

The 1-5 scoring system below provides a simple and consistent way to evaluate each criterion. Each score will be supported by a written justification in the individual reviews, ensuring the evaluation produces a meaningful analysis rather than just numerical results.

Scoring Rubric

Score	Meaning	Guidance
1 (Poor)	Major issues	Feature is missing or very badly implemented
2 (Fair)	Lots of issues	Works but causes user frustration or confusion
3 (Good)	Acceptable	Meets basic expectations but has some issues
4 (Very good)	Strong	Works well with small improvements needed
5 (Excellent)	Best	Fully working order with no noticeable issues

Scoring Table (For reviews below)

Topic	Focus	Criteria	News	Entertainment	Sport
Usability	Consistency, learnability, error prevention (Nielsen heuristics)	Ease of use & efficiency	4/5	4/5	4/5
Accessibility	Contrast, alt text, keyboard access (WCAG principles)	Inclusive design	4/5	4/5	5/5

Visual Design & Hierarchy	Clear structure, typography, visual priority	Layout & readability	4/5	5/5	4/5
Colour Theory	Contrast, consistency, emotional impact	Use of colour	5/5	4/5	5/5
Semiotics	Clarity of icons, symbols, and visual cues	Meaning of visuals	5/5	5/5	3/5
Navigation	Clear menus, logical flow, ease of movement	Site structure	4/5	5/5	4/5
User-Centred Design	Meets user needs, supports task completion	User focus	4/5	4/5	3/5
Cognitive load	Low cognitive load, recognition over recall	Mental processing	3/5	3/5	4/5
Interaction & Feedback	Feedback, responsiveness, clear error messages	System response	4/5	5/5	4/5
Ethical Design	Transparency, no misleading content	Honest interaction	4/5	4/5	5/5
Dark Patterns & Persuasive Design	Avoids manipulation, uses ethical persuasion	Behaviour influence	3/5	2/5	4/5
Privacy & Security	User consent, secure data practices	Data handling	3/5	4/5	3/5
User Support	Help systems, guidance, error recovery	Assistance features	2/5	3/5	2/5

Individual Contributions

B01021275 Kaden Hunter - Individual appendix - BBC News Website evaluation

Introduction - This individual appendix applies the evaluation framework to the BBC News Website <https://www.bbc.co.uk/news>. BBC News is one of the most visited news websites globally and is operated by the British Broadcasting Corporation, who are a public service broadcaster. As a publicly funded organisation the BBC is expected to maintain high standards of usability, accessibility and user trust. This evaluation assesses the website against all the framework criteria using HCI and design principles. Each section is scored from 1-5 based on the table above and supported by the written justification through my report.

Usability (4/5) - BBC News performs well against Nielsen's ten usability heuristics. The navigation bar, header, logo, and overall page layout remain consistent across the site, so users do not need to relearn the interface when moving between sections. The BBC logo acts as a reliable home link, supporting predictable navigation, article pages follow a clear repeated structure with a headline, date, hero image, and body text, reducing cognitive effort for readers. Error prevention is also effective, as failed search queries provide clear messages and alternative suggestions rather than broken pages. A weakness is that live blog updates and auto refreshing content do not always clearly show when content has changed which affects visibility of system status [2]. Overall, BBC News provides an efficient and easily learnable experience for users.

Accessibility (4/5) - The BBC has invested a lot in accessible design across its digital services [12]. Keyboard navigation works throughout the website, a skip to content link is visible when tab focus is activated and images include descriptive alt text supporting screen reader users. The website uses semantic HTML which helps screen readers to interpret page structure, colour contrast meets the WCAG for primary text, but some secondary elements like timestamps with grey text may cause difficulty for users with any form of visual impairment. Auto-generated captions on live videos are inaccurate quite a lot which could affect hearing impaired users. Ensuring all the secondary text meets the WCAG guidelines would make the BBC's website more accessible overall.

Visual design and hierarchy (4/5) - BBC News has a clean and structured visual design, the homepage uses a card-based grid with a full width hero story at the top which guides the user's eye from the most important content downward which creates a clear visual hierarchy [5]. Typography is consistent throughout as well, headlines use bold text at a large size, and body text is set at a comfortable reading size with good line spacing. The way each card groups the headline, summary and image together follows Gestalt's principles of proximity and similarity [5]. The main weakness is that the homepage is visually dense during major news events where many big stories are being shown at the same time which can reduce a good sense of visual hierarchy.

Colour theory (5/5) - BBC News uses a minimal colour palette of black, white and red. Red is used only for breaking news labels which draws on its understood association of urgency and importance [3]. Breaking news is communicated with both a red label and

the word 'BREAKING', to ensure that colour isn't the sole means of conveying importance. The restrained use of red means it always retains its urgency signal and never just becomes decorative. White and black backgrounds provide the best contrast for the body text. The use of colour across BBC News has great consistency, purpose and is fully accessible.

Semiotics (5/5) - Icons across BBC News follow established website conventions and are immediately recognised to most users. For example, the magnifying glass shows search and the hamburger icon opens the navigation menu on mobile. For every navigation icon there is also text labels meaning that less experienced users who may not recognise the icons will still be able to identify what they are.

Navigation and information architecture (4/5) - The two-level navigation structure keeps primary categories of the website uncluttered whilst still offering access to a wide range of content. For example, within the News section, a secondary bar allows the user to filter by topic areas such as the UK, World or Culture. Search is accessible from every page and there are breadcrumb trails on article pages to help users understand their position in the site structure. A weakness is the 'More' dropdown in the primary navigation which adds a few more links to the page which could possibly slow down user decision making. Restructuring the menu just to the most accessed sections could improve the navigation on the website more.

User-centred design (4/5) - BBC News has very readable font sizes, article pages are free from intrusive advertising, and the reading experience is clean and focused. The sign-in personalisation feature allows users to follow specific topics and save articles which supports goal-directed browsing. However, sign-in prompts are sometimes persistent and sometimes makes users feel that they need an account to access content when they don't, which could annoy users who just want to read without registering. More transparent communication of the optional choice of personalisation would better help users.

Cognitive load (3/5) - The homepage presents a great volume at the same time, during major news events there could be higher than 30 story cards alongside promotional banners, topic navigation, weather widgets and video content within the first two scrollable screens which can create a very high cognitive load for users as most things does not contribute to the users goal [6]. Sweller's research shows that this overloads working memory and leads to frustration and slower task completion. BBC news helps with this somewhat through visual hierarchy, but the volume of content remains a strong weakness especially on mobile.

Interaction and feedback (4/5) - Interactive elements respond well across BBC news, links and story cards change appearance on hover, video controls are clear and 404 pages are easy to understand. Don Norman identifies feedback as a key design principle and BBC News provides strong visual feedback through these [3]. However live blog auto refreshing can add new content at the top of the page without clearly notifying users which conflicts with Shneiderman's principle of information feedback

[8]. Some content heavy pages also lack visible loading indicators which leaves users unsure if content is loading.

Ethical design (4/5) - BBC News avoids most unethical design choices such as clickbait, autoplay audio and disguised advertisements. Sponsored content is clearly labelled and news, analysis and opinion are separated well. The main issue is the repeated account prompts that can make sign in seem required when it is optional. The ACM Code of Ethics highlights transparency and avoiding harm [9], and clearer account messaging would improve trust.

Dark Patterns and Persuasive design (3/5) - BBC News has no paywalls, false urgency or countdown timers which makes it cleaner than most commercial news sites. But the cookie banner shows 'Accept all' more prominently whilst rejecting cookies requires extra steps. This is classified as a dark pattern [10]. ICO guidelines states that consent should be as easy to refuse as to accept which is not fully met here [11].

Privacy and security (3/5) - BBC News uses HTTPS throughout and links its privacy policy in the footer however the full privacy policy is written in a more complex language making it harder for most users to understand therefore resulting in most skipping over it. The cookie consent issue also weakens informed consent as UK GDPR requires consent to be clear and easy to withdraw [11].

User support and help systems (2/5) - User support is the weakest area of BBC News, there is no visible help link, FAQ, or in page guidance. Nielsen states that help should be easy to find and task focused [2], but BBC support is placed on a separate site with little indication on where to find it. If a user faces issues such as videos not loading there is no clear support available which can cause frustration.

Recommendations for improvement - I have recommendations for improvement based from some low scoring areas of the evaluation, the first being that the cookie consent banner should include a clearly labelled 'Reject all' button on the first screen with the same visual weight as 'Accept all', removing the need to open additional settings which would better meet ICO guidance on valid consent [11]. Second, during busy periods the homepage should display fewer stories at once by using expandable topic sections which allow the users to open the areas they are most interested in instead of being faced with too much content at one time. This would reduce cognitive load, especially on mobile devices and improve the overall reading experience [6]. Third, BBC news should provide stronger in-site user support through a visible help link in the footer that directs users to a BBC news specific FAQ section for common issues like video playback, account access and loading problems. Adding guidance for features such as personalisation would also better support Nielsen's principle that states that help should be easy to find and available at the point of need [2].

Overall BBC News performs well across most of the framework criteria, showing a strong commitment to usability, accessibility and ethical design as expected of a public news broadcasting service.

Individual Contributions

B01022158 Ellie Bradshaw - Individual appendix – YouTube evaluation

Introduction- I will be evaluating <https://www.youtube.com> by using the framework to see how well it supports user needs. YouTube is one of the most popular entertainment websites globally. YouTube is used daily by many people and because of this there is an expectation for it to have good usability and accessibility so it can be used easily by everyone. I will be covering all points mentioned in the framework and giving it a score based on the table, this will determine whether YouTube is user friendly or not.

Usability (4/5) – YouTube has consistent navigation throughout its website, the YouTube logo is a home button no matter what page you are on, there is also a folded navigation bar in the top left corner that will also allow users to easily navigate, they can click on the home button, or go to different features YouTube offers easily and quickly. The search bar also stays present throughout all pages; this will allow a user to search for things they need without having to click off the video they are currently watching. It also does well for error prevention, an example of this is when searching it auto fills your request to what it thinks you're searching for, this will reduce spelling errors and wrong results. There is also a clear way to reverse something you've done, if you click the home button it takes you back to the last thing you searched meaning if you click on the wrong video, it is easy to return and find the correct one. However, YouTube does have an autoplay feature that can be difficult to exit out of if the user doesn't wish to watch that video. These all follow Nielsen's ten usability heuristics and Schneiderman's 8 rules.

Accessibility (4/5)- All the text in YouTube is large and clear lettering that will be easily read by users as WCAG states there should be text alternatives, YouTube does this by using symbols, the home button has a symbol of a house along with the word below it and the search bar has a magnifying glass. When looking through videos when your mouse is hovering over a video a blue box will appear showing users what video they will be clicking on, allowing it to be more accessible for some people, however this is only for the videos and doesn't apply to the navigation bar or home button. YouTube also provides subtitles for almost all videos, this is great for people who are hard of hearing, although the subtitles aren't always accurate and it depends on how clear the sound is on the actual video.

Visual design and hierarchy (5/5)- Throughout the website YouTube has a clear visual hierarchy the thumbnails for the clips are bright and large drawing your attention before the text below, this will help users when deciding what they want to watch. There is a consistent layout with the videos, there's a thumbnail, the name, the channel and views, this will reduce cognitive overload as everything is in the same place, following Gestalt principles. The videos are also evenly spaced which also supports Gestalt's principles. The typography is readable, with title and important information being in large bold letters. The sections are clearly labeled and grouped together which will allow users to scan quickly and decide on what one they want.

Colour theory (4/5)- YouTube has a very minimalistic colour palette, it mainly uses red, white and black, red is used to highlight important things like subscribe or in its logo, red is a striking colour and will draw users' attention easily. Black is used for text and in the

navigation bar and white is the background colour. It also used blue to show the sign in button and when hovering over videos this will draw users' attention to these showing their importance. However, the thumbnails are often very colourful which can become overwhelming and they don't fit the colour palette of the rest of the website.

Semiotics (5/5)- YouTube uses icons throughout its design. Some of the most common are the thumbs up and thumbs down buttons, the search magnifying glass and bell for notifications. These symbols are known world-wide and can help users when navigating the website. It uses a pairing of both text and symbols like on the navigation bar each symbol is accompanied by the name for example under the YouTube shorts symbol it says shorts, allowing users to recognise and remember what different symbols mean.

Navigation and information architecture (5/5)- There is a folded navigation bar at the top left corner that users can click at any time and see all the different options YouTube has, for example history or home, these are also grouped by category as well. Along the top is also has the different categories YouTube offers like, music or gaming, these will change depending on what the user searched for. There is a side along the left-hand side that shows the four main sections users would need, Home, shorts, subscriptions and you. This will help users easily navigate the website. This matched Nielsen's usability heuristics of using a clear organised structure.

User-centred design (4/5)- YouTube has a very user centred design, it has videos that it recommends, determined by what users have previously watched. There are different playback controls in a video allowing users to choose the best way for them to watch. By signing into your account, you are able to access all your preferences and data on multiple accounts. The first thing users do when they open YouTube is search for what they want this gives users full control over what they are watching, this agrees with the Nielsen usability heuristics. However, as YouTube is very personalised it may mean users never branch outside of what they normally watch and never get to see new content, this can also be repetitive and become boring for users as all recommended videos are very similar.

Cognitive load (3/5)- YouTube follows the same layout for every video and uses the same controls to exit and same home button for every page. Everything in it is grouped into sections which are easy to read and scan, this all follows Gestalt's principles. However, YouTube can still feel very overwhelming for users. There are lots of thumbnails which are brightly coloured and grab users' attention, the preview of the video also plays when you hover on it, this can overload them. And there seems to be an endless supply of videos on a certain topic which can also feel overwhelming for users.

Interaction and feedback (5/5)- As soon as a video loads along the bottom it shows the user where the controls for the video is and will show up when the user moves the mouse onto the video. While the video is playing users can see how far they've got in the video and how much time is left by looking at the progress bar along the bottom, following the Nielsen heuristics. When the like or dislike button is pressed it gives the user instant feedback that the action has been submitted with the like count going up and the icon changing colour. Before a new video plays there is a countdown showing allowing the users time to decide if they want to watch the next video. YouTube gives

the users visual feedback throughout the website which is following Schneiderman's 8 rules.

Ethical design (4/5)- YouTube allows users to report videos they feel they need to if it has disrespectful or harmful content. It also makes sure to ads are labelled as ads which will ensure users know they are ads even if it is not obvious. There is also an alternative website called "YouTube kids" this deviant of the original YouTube website is meant for kids which means all content on it is safe guarded and meant to be viewed by children, meaning children can't see harmful videos. However, YouTube focuses more on engagement that user wellbeing meaning they will continue to show videos users watch, disregarding if they are harmful or not.

Dark Patterns and persuasive design (2/5)- YouTube does have a very persuasive design. The videos that you see a recommended for you based on what you have watched before encouraging users to watch more videos than they intended when watching the app. The auto play feature also encourages this; it created an endless cycle that users can get lost it as every time they finish a video another one is showed that they will most likely enjoy. The YouTube shorts is also a very persuasive and addictive feature as users are likely to scroll for hours not realising as the videos are so short. Some dark patters that YouTube has are ads that resemble normal videos, while they are labelled user could miss this and become mislead. The continue watching section will also encourage users to finish videos even if they turned it off intentionally.

Privacy and security (4/5)- Users can easily manage their privacy and security in the YouTube website, they can manage their watch history, search history, ad personalisation and data collection. As required by WCAG YouTube uses encrypted connections which will protect users' data. Users can also restrict what videos can be shown, this can be good for young viewers to stop them seeing inappropriate or harmful content. However, using user data YouTube shows them targeted ads specifically for then which users could find intrusive, and users must change the default data collection controls and manually stop YouTube from doing this.

User support and help systems (3/5)- If users are having problems there is a dedicated support centre in google support, this lets users describe a problem or pick from regular problems other users face. When hovering over buttons it tells the users what they do, providing support and help, the error messages are also clear, "check connection" or "this video is unavailable." While there is lots of help it is not built into the website, users will have to leave the sight to gain more information on their problem. There is an error message, but this doesn't always explain how to fix the issue.

Recommendations for improvement- Some improvements I feel YouTube could make would be to have videos in different collapsable sections, this will reduce cognitive load on the home page and help users to navigate the website. YouTube could also give users the option to stop auto play, this can stop users seeing unwanted videos. They should also integrate more help into the website itself, they could do through adding ? beside icons with a brief explanation on what it is and how it works. Also change some defaults in setting like personalised ads, make these privacy setting clear to users and give them to option to turn them on instead of having them on by default. This will improve privacy and security on the website.

Introduction to Human Computer Interaction – CW2

B00988940 Rian O’Kane Individual Appendix Sky Sports

Introduction- My individual appendix is based on the Sky Sports website [Sky Sports - Sports News, Transfers, Scores | Watch Live Sport](#). Sky Sports is one of the most popular sports websites in all of Europe and is most used by people to view the latest headlines and videos within the sporting world. The website is expected to follow HCI and UX principles in their website so I’ll be assessing the website based on this criteria and I’ll be ranking them based on a scale of 1-5.

Usability 4/5 - On the Sky Sport Website, the usability is generally pretty good with a few minor issues that can hinder the user experience. On the front home screen, the header will include the home and sports option, and these will allow the user to return to the home screen and to find the sports news that they are looking for. The logo on the top left of the screen will also make it easier for the user to return to the home screen. On the home screen it will include a variety of different headlines that happening throughout the day. However, this means that the home page can be quite clutter for some people which means that some people might struggle to get through the home screen.

Accessibility 5/5 - Sky Sports makes their website accessible for users with disabilities by aiming to meet the AA level of the Web Content Accessibility Guidelines 2.1. They also follow Ofcom guidelines and work with groups including the Royal National Institute of Blind People and Action on Hearing Loss to improve accessibility for users.

Visual Design and Hierarchy 4/5 - The website uses a digital first design approach to help emphasizes the importance of hierarchy within the webpage. The websites use a typographical design by using a bespoke font and this will allow for the flexibility across the content from the headlines to the stats. It follows readability to help ensure that it is readable and accessible for the user. Throughout the website it will use images to highlight an important moment in a match or event thus it will be catching the user’s attention. Despite that I believe that the website has some issues when it comes to homepage. Because there is so much being put onto the homepage it can lead to the user be coming overwhelmed.

Colour Theory and Use of Colour (5/5)- There are four colours that are used on the Sky Sports website. These are blue, red, black and white. White is used in the background, and this will help make the headline stand out more to the user which would normally be in black. There is blue used in don’t miss this column of the homepage and this will help it to catch the user’s attention. Whenever there is a latest new story heading on to the webpage it would normally have a yellow background with the headline being in black. Red is used when something is live on the channel and this will catch the user attention.

Semiotics 3/5- The semiotics on the Sky sports website helps the user achieve certain goals on the website. The play button on each of the main videos allows for the user to be able to watch videos on the website. The arrows on the side of the don't miss this part allows the user to see more videos on the website. The icon on the top right of the page will allow the user to log in to their sky account. However, I feel that there is an issue or two on the website. On the live section the start button is much smaller and thus it will be harder to start a video. I also feel that some of the play buttons are render mute as the user can simply click on the video they want to watch.

Navigation and Information architecture 4/5 - Navigation will help make sure that the user will find what they're looking for. On the top navigation bar, it will allow the user to find the news on the sports they like. If the user wants to find the top headlines throughout the day it will all be presented on the homepage for people to find. This intern will help keep the user up to date with all the latest news through the day. All these different components will help to keep people from getting confused on the website and help them find what news story they're looking for. My only issue is that two of the sky bet and more section will include links to other websites and this could lead to the user getting confused.

User-Centred Design 3/5 – Sky Sports makes sure that the user can tailor what they view on the website based on their preferred sports. They make sure that their headlines include big black writing, and they make sure that the main headline is bigger than the others. These two factors are crucial to help to catch the user attention and follow the principles of UC Design as stated in Medium's ten essentials' principles for user centred design. However, towards the bottom of the home page there is a column for Sky news and Sky bet and these can throw users off as they have nothing to do with sports.

Cognitive load 4/5 - The front bar on the home page will help to spread all the headlines that they want to see. The homepage is layout in different column including latest news and watch and this will allow the user to find what they're looking for much easier to find what they're looking for. The layout of the homepage will help the user make their decision on what story they want to view according to DigitalHype. However, despite saying that I do believe that the homepage can be quite cluttered as they are plenty of headlines that are being crammed into the homepage thus it will make it harder for the user to decide what to view.

Interaction and feedback 4/5 – On Sky Sports it is very consistent with different interactive elements. The webpage is very consistent when it comes to displaying both videos and headlines when they are clicked on. The webpage is also good at making their headlines easy for people to read throughout the font size. Thus, it will follow Ben Shneiderman rules of principle design through consistently which is followed greatly by Sky Sports. But the main page can often change headlines when the users refresh the page thus the user might miss out on a story that they might want to read.

Ethical design 5/5 – Sky Sports follows the ACM Code of Ethics by clearly separating websites such as Sky News and Sky Bet into their own sections. The website avoids misleading design choices and includes a privacy centre that allows users to manage their privacy settings.

Dark Patterns and Persuasive design 4/5 - Sky sports don't have a lot of Dark Patterns and Persuasive design choices. The adverts will take you to the website, it doesn't misdirect the user onto another website, and the website is easy to sign up to without the worry of being unable to cancel as the website is free to use and without pay. The website also doesn't trick the user into sharing personal information with them thus making the user trust them more.

Privacy and security 3/5 - HTTPS is used throughout Sky sports and privacy is in its own column at the bottom of the homepage called Privacy and Cookies notice. Once I clicked onto the column, I noticed an issue with the privacy. If you have any questions about your right to privacy it won't always give you a straight answer in some cases and thus it can lead to a tedious process where you must get in touch with Sky to get your questions answered which can take up a lot of time.

User support and help systems 2/5 - At the bottom of the website page there is an option to get help with any issues through the contact us option. This will follow Nielsen's heuristics as it will make it easy for the user to find. However, when you enter that option, it will give a lot of questions for different parts of the Sky account including sky bet and NOW TV and this means that the user won't be able to get anything solved because they can't make their questions to Sky Sports.

Recommendations for improvement- Based on what I had reviewed there are a few improvements that could be made to Sky Sports webpage. The first of these improvements would be with the clutter home page. I believe that in the main page should remove the far-right advertisement to allow for more room in the home page. I would also remove the sky bet and sky news columns on the homepage so that the page can be less cramped and make it easy for the users to find what they're looking for. I would make sure that the user is able to get their questions answered in a timely fashion and that the help column is tied to the Sky Sports website to prevent confusion. Another improvement that I would make sure that all the buttons and play buttons on the live section would serve their functions. This would help make sure that everything is able to meet the standards of semiotics studies.

Overall, I believe that Sky Sports does a good job in following HCI and UX principles to help them ensure that the website can meet the standards of the user and to help it run properly.

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